



Air University, Aerospace and Aviation Campus, Kamra

Department of Computer Science

BS. Cyber Security

**Project Title: Information Assurance planning for
FalconPetro Marketing Ltd.**

BSCYS – 3B

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ABSTRACT

FalconPetro Marketing Ltd. is a nationwide oil marketing company responsible for the procurement, storage, distribution, and retail sale of petroleum products through more than 350 fuel stations. Due to its heavy reliance on information systems such as ERP, supply chain databases, and POS platforms, FalconPetro faces significant information assurance risks that can directly impact national fuel availability, financial integrity, and public safety.

This project identifies and analyzes FalconPetro's critical information, physical, software, and human assets, with particular focus on its supply chain and ERP systems. Major risks identified include ransomware attacks on core systems, phishing-based credential compromise, POS outages, insider manipulation of pricing or inventory data, third-party access vulnerabilities, and physical incidents at fuel depots.

An Information Assurance (IA) plan was developed incorporating security policies, access controls, incident response procedures, business continuity planning, and disaster recovery strategies. The project demonstrates how coordinated governance across executive management, CISO, CIO, COO, and HSE leadership enables operational resilience. The key outcome of the IA plan is a layered defense approach that integrates cybersecurity controls with operational safety and continuity measures, ensuring uninterrupted fuel supply, regulatory compliance, and protection of critical national infrastructure.

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2. Introduction

Background of Information Assurance

Information Assurance (IA) focuses on protecting the confidentiality, integrity, and availability (CIA) of information systems. In critical infrastructure sectors such as oil and gas marketing, failures in IA can result in financial loss, public disruption, environmental damage, and safety hazards. Modern IA extends beyond cybersecurity to include operational continuity, governance, and risk management.

Purpose of the Project

The purpose of this project is to design and evaluate an Information Assurance framework for FalconPetro Marketing Ltd., identifying its assets, risks, and control measures to ensure secure and resilient fuel distribution operations.

Scope and Limitations

This report covers information security, risk assessment, business continuity, and disaster recovery for FalconPetro's IT enabled operations. It does not include detailed software coding, penetration testing, or legal enforcement mechanisms beyond policy-level analysis.

3. Company Selection

Company Type

Oil Marketing Company (OMC)

Company Size

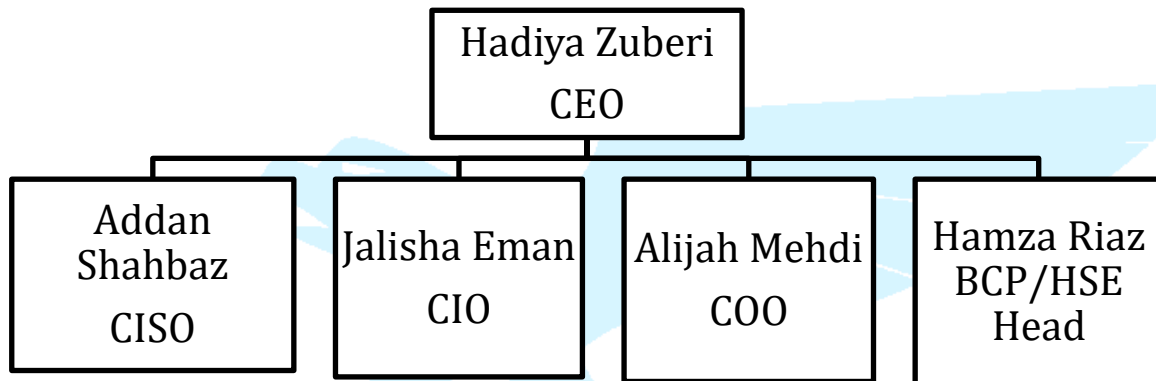
- Nationwide operations
- 350+ fuel stations
- 6 regional depots
- Approximately 900 employees

Company Description

FalconPetro Marketing Ltd. procures petroleum products from refineries and distributes them to retail fuel stations through a centrally managed logistics and IT infrastructure. Operations rely heavily on ERP systems, supply chain databases, POS platforms, and third-party transporters.

4. Upper Management and Governance Structure

Organizational Chart:



Information Assurance Policies & Procedures

1. Information Security Policy

Policy Statement

FalconPetro Marketing Ltd. is committed to protecting the confidentiality, integrity, and availability (CIA) of all information assets that support fuel procurement, storage, distribution, and retail operations. Information security is a mandatory responsibility for all employees, contractors, and third parties.

Procedures

1. Information security controls shall be designed and maintained in alignment with ISO/IEC 27001 principles.
2. All information assets shall be classified and protected according to their business criticality.
3. Security incidents must be reported to the IT Security team within **30 minutes** of detection.
4. Annual internal security audits shall be conducted, with findings tracked to closure.

2. Access Control & Identity Management Policy

Policy Statement

Access to FalconPetro systems shall be restricted to authorized users only, based on the principle of least privilege and segregation of duties.

Procedures

1. All users shall be assigned unique user IDs; shared accounts are prohibited.
2. Multi-factor authentication (MFA) shall be enforced for ERP, POS, VPN, and administrative systems.
3. Access rights shall be reviewed **quarterly** by system owners and approved by department heads.
4. User access shall be revoked immediately upon employee termination, role change, or vendor contract expiry.

3. Acceptable Use Policy

Policy Statement

FalconPetro IT resources are provided strictly for authorized business use to support safe and reliable fuel operations.

Procedures

1. Installation of unauthorized software or hardware is prohibited.
2. Email and internet usage may be monitored for security and compliance purposes.
3. IT systems shall not be used for illegal activities or personal commercial gain.

4. Data Classification & Protection Policy

Policy Statement

All data shall be classified and protected according to its sensitivity and business impact.

Data Classification

- **Confidential:** Pricing data, payroll, contracts, ERP financial records
- **Internal:** Operational and logistics reports
- **Public:** Marketing and publicly released information

Procedures

1. Confidential data must be encrypted at rest and in transit.

2. Data sharing with third parties requires documented management approval.
3. Daily operational and sales data shall be securely backed up in accordance with the Backup & Recovery Policy.

5. ERP & Core Systems Security Policy

Policy Statement

The SAP ERP system and supply chain databases are Tier-1 critical assets and must operate securely and reliably at all times.

Procedures

1. ERP access and activity shall be logged and continuously monitored.
2. Any changes to pricing, inventory, or financial data shall require **dual authorization**.
3. ERP and database backups shall be performed daily and stored at an offsite Disaster Recovery location.
4. Privileged access shall be reviewed monthly to prevent misuse or privilege creep.

6. POS & Payment Systems Policy

Policy Statement

POS systems and payment gateways must ensure secure, uninterrupted, and compliant customer transactions across all fuel stations.

Procedures

1. POS terminals shall operate only approved and patched software.
2. Payment data shall comply with PCI-DSS requirements.
3. POS systems shall support offline mode during central system outages.
4. Any POS outage must be reported immediately to IT Operations.

7. Incident Response Policy

Policy Statement

FalconPetro shall respond swiftly to security and operational incidents to minimize business disruption and safety risks.

Procedures

1. Incidents shall be classified as Low, Medium, High, or Critical.

2. Ransomware or ERP compromise shall trigger immediate Disaster Recovery procedures.
3. Incident reports shall be documented within **24 hours** using a standardized reporting template.
4. Post-incident reviews shall be conducted to identify root causes and corrective actions.

8. Business Continuity & Disaster Recovery Policy

Policy Statement

FalconPetro shall ensure continuity of fuel supply and critical IT services during cyber, physical, or natural disruptions.

Procedures

1. A Disaster Recovery site shall be maintained and tested **quarterly**.
2. All Tier-1 systems shall have defined Recovery Time Objectives (RTO) and Recovery Point Objectives (RPO).
3. Fuel stations shall maintain documented manual fallback procedures during IT outages.
4. Cyber incidents shall be integrated into Business Continuity and HSE drills.

9. Backup & Recovery Policy

Policy Statement

All critical information assets shall be backed up to prevent data loss and ensure rapid recovery.

Procedures

1. Daily automated backups shall be performed for ERP, POS, and supply chain systems.
2. Weekly full backups shall be stored securely offsite.
3. Backup restoration tests shall be conducted **monthly** for Tier-1 systems.

10. Vendor & Third-Party Access Policy

Policy Statement

Third-party access shall be controlled to protect FalconPetro systems and data.

Procedures

1. Vendors shall sign confidentiality and security agreements before access is granted.
2. VPN access shall be time-bound and role-restricted.
3. Third-party activity logs shall be reviewed **weekly** for anomalies.

4. Vendor access shall be suspended immediately during security incidents.

11. Insider Threat & Fraud Prevention Policy

Policy Statement

FalconPetro shall prevent misuse of systems, data, and fuel assets by insiders.

Procedures

1. Segregation of duties shall be enforced for pricing, inventory, and approvals.
2. Audit logs shall be reviewed weekly.
3. A confidential whistleblower mechanism shall be maintained.

12. Physical Security Policy

Policy Statement

Physical assets, including fuel depots, stations, and data centers, shall be protected against unauthorized access and hazards.

Procedures

1. CCTV surveillance shall be maintained at depots and fuel stations.
2. Access badges shall be required for data center entry.
3. Fire and emergency drills shall be conducted annually.

13. Operational Safety & Fuel Handling Policy

Policy Statement

Safe fuel handling is mandatory to protect human life, the environment, and company assets.

Procedures

1. Staff shall receive regular fire safety and spill response training.
2. Tanker movements shall be GPS-tracked and logged.
3. Fuel theft or safety incidents shall be reported immediately.

14. Power & Infrastructure Resilience Policy

Policy Statement

Critical operations shall be protected from power and infrastructure failures.

Procedures

1. UPS and generators shall be maintained at HQ, depots, and data centers.
2. POS terminals shall support offline operations.
3. Preventive maintenance schedules shall be documented and reviewed

15. Compliance & Audit Policy

Policy Statement

FalconPetro shall comply with all applicable laws, regulations, and industry standards.

Procedures

1. Annual internal audits shall be conducted.
2. Regulatory compliance reviews shall be documented.
3. Non-compliance issues shall be tracked until resolved.

16. Security Awareness & Training Policy

Policy Statement

Employees shall be continuously educated on cybersecurity and operational risks.

Procedures

1. Cybersecurity awareness training shall be conducted **quarterly**.
2. Phishing simulations shall be performed regularly.
3. Cyber scenarios shall be included in operational safety drills.

Critical Business Functions

1. Fuel Procurement & Pricing Management

Function Description

Responsible for procuring petroleum products, setting fuel prices, managing margins, and ensuring regulatory compliance.

Supporting Assets

- SAP ERP System
- Financial Records & Pricing Databases
- Executive Approval Workflows

Impact of Disruption

Incorrect pricing or procurement delays can lead to financial losses, regulatory penalties, and market instability.

2. Supply Chain & Distribution Operations**Function Description**

Manages fuel movement from refineries to depots and from depots to petrol pumps, including tanker scheduling and route planning.

Supporting Assets

- Supply Chain & Distribution Database
- Logistics Scheduling Systems
- GPS Tracking Systems

Impact of Disruption

Failure results in fuel shortages, station outages, and loss of public trust.

3. Retail Fuel Sales & Payment Processing**Function Description**

Ensures uninterrupted sale of fuel to customers and secure processing of cash, card, and digital payments at fuel stations.

Supporting Assets

- POS Systems
- Payment Gateways
- Network Connectivity

Impact of Disruption

Direct and immediate revenue loss and customer dissatisfaction.

4. Financial Management & Regulatory Reporting

Function Description

Handles invoicing, taxation, payroll, financial reconciliation, and compliance with regulatory reporting requirements.

Supporting Assets

- ERP Financial Modules
- Audit & Compliance Records
- Secure Databases

Impact of Disruption

Regulatory non-compliance, delayed payments, legal penalties, and reputational damage.

5. Information Systems Availability & Integrity

Function Description

Ensures availability, integrity, and reliability of ERP, POS, databases, and communication systems that support nationwide operations.

Supporting Assets

- Data Centers & DR Site
- Backup & Recovery Systems
- Network Infrastructure

Impact of Disruption

Complete operational shutdown across fuel distribution and retail operations.

6. Cybersecurity & Risk Management

Function Description

Protects information assets from cyber threats, insider misuse, and third-party risks through monitoring, policies, and incident response.

Supporting Assets

- Security Monitoring Tools
- Access Control Systems
- Incident Response Procedures

Impact of Disruption

Data breaches, ransomware attacks, operational paralysis, and loss of public trust.

7. Business Continuity & Disaster Recovery

Function Description

Ensures continuation of fuel supply and IT services during cyber incidents, natural disasters, or physical emergencies.

Supporting Assets

- Disaster Recovery Site
- Business Continuity Plans
- Manual Fallback Procedures

Impact of Disruption

Extended downtime, fuel shortages, and safety risks.

8. Operational Safety & Environmental Protection

Function Description

Protects employees, customers, and the environment through safe fuel handling, fire prevention, and spill response.

Supporting Assets

- Fuel Depots & Stations
- Safety Training Programs
- Emergency Response Plans

Impact of Disruption

Loss of life, environmental damage, legal penalties, and shutdowns.

9. Vendor & Third Party Management

Function Description

Manages relationships with transporters, payment providers, and IT vendors to ensure secure and reliable operations.

Supporting Assets

- Vendor Contracts
- VPN & Access Logs

- Third-Party Portals

Impact of Disruption

Supply chain compromise, data leakage, or service interruptions.

5. Asset Identification

5.1 Information Assets

Supply Chain & Distribution Database

Records fuel quantities, depot-to-pump allocation, tanker schedules, delivery routes, and dispatch confirmations.

Petrol Pump Transaction & Payment Records

Contains daily sales, settlements, outstanding balances, and reconciliation data for each fuel station.

Financial Records

Includes invoices, tax calculations, pricing records, profit margins, and regulatory financial reports.

Audit & Compliance Records of Petrol Pumps

Inspection reports, safety audits, quality checks, and regulatory compliance documentation.

Vendor & Transporter Information

Contracts, delivery agreements, third-party access details, and performance records.

Business Continuity & Incident Records

Past incident logs, recovery plans, emergency contact lists.

5.2 Physical Assets

Fuel Depots & Storage Facilities

Physical storage of petroleum products prior to distribution.

Fuel Tankers & Delivery Vehicles

Vehicles responsible for transporting fuel from depots to petrol pumps.

Petrol Pumps & Fuel Stations

Retail outlets selling fuel to customers.

Servers

Host supply chain databases, ERP systems, and internal applications.

Network Devices

Routers, switches, firewalls connecting headquarters, depots, and fuel stations.

Data Centers / Server Rooms

Physical locations housing core IT infrastructure.

Power Backup Systems (UPS, Generators)

Ensure availability of critical systems during power outages.

5.3 Software Assets**ERP System**

Manages inventory, pricing, invoicing, financial reporting, and integration with petrol pump operations.

Supply Chain Management System / Logistics Scheduling Tool

Generates tanker routes, delivery schedules, and dispatch plans.

Central Databases

Store supply chain data, payment records, audits, and operational information.

POS Software at Petrol Pumps

Processes customer payments and synchronizes sales data with central systems.

Email & Collaboration Systems

Used for internal communication between headquarters, depots, and fuel stations.

Vendor & Transporter Portals

Provide controlled access to third-party partners.

Backup & Recovery Software

Used for data backup, restoration, and disaster recovery.

5.4 Human Assets**Executive Management**

CEO, CIO, CISO, COO responsible for strategic decisions, policies, and crisis management.

Supply Chain & Operations Staff

Personnel managing dispatch, logistics, depot operations, and fuel distribution.

System Administrators & Database Administrators

Maintain ERP systems, databases, servers, and access controls.

Security & Information Assurance Staff

Responsible for information security policies, monitoring, and incident response.

Petrol Pump Staff & Managers

Operate fuel stations and handle daily customer transactions.

Third-Party Transporters & Vendors

External parties involved in fuel transportation and logistics

6. Risk Assessment**Risk Assessment Method:**

Risk is assessed using:

$\text{Risk} = \text{Likelihood} \times \text{Impact}$

Major Risks Identified:**1. Phishing or cyberattack leading to ERP compromise.**

The main vulnerability is employee exposure to phishing emails and shared system access. The likelihood is **High** due to frequent phishing attempts, and the impact is High because ERP controls fuel procurement, pricing, and payroll. Overall risk level is **High**.

2. Ransomware attack on HQ or data center systems.

Vulnerabilities include centralized data center, limited segmentation, and delayed patching. Likelihood is **Medium**, but impact is Very High as it can shut down operations nationwide. Overall risk level is **High**.

3. POS system outage at fuel stations.

This can occur due to network failures, software bugs, or power issues. Likelihood is **Medium**, impact is High because fuel sales stop immediately. Overall risk level is **Medium to High**.

4. Payment gateway failure affecting card and mobile payments.

Vulnerability exists due to dependency on third-party payment providers. Likelihood is **Medium**, impact is High because customers cannot pay and sales are disrupted. Overall risk is **Medium**.

5. Insider manipulation of fuel pricing or inventory data.

Vulnerabilities include privileged user access and weak monitoring. Likelihood is **Low to Medium**, impact is High due to financial loss and reputational damage. Overall risk is **Medium**.

6. Fuel station fire or depot accident.

Vulnerabilities include fuel handling risks and safety non-compliance. Likelihood is **Low**, but impact is Very High due to loss of life, assets, and regulatory action. Overall risk is **Medium to High**.

7. Power outages or natural disasters affecting depots.

Vulnerability comes from reliance on grid power and regional exposure. Likelihood is **Medium**, impact is High due to supply disruption. Overall risk is **Medium**.

7. Recovery & Disaster Recovery Plan

Who We Are & Why Recovery Matters

At FalconPetro, we are a nationwide oil marketing company operating more than 350 fuel stations. Our operations—including fuel supply, pricing, payment processing, and logistics—are fully dependent on our IT systems. Any disruption to these systems immediately affects fuel distribution, prevents customers from making payments, and halts station operations.

As the Chief Information Officer (CIO), it is my primary responsibility to ensure that all critical IT systems are restored quickly and safely whenever any disruption occurs. My role is central to maintaining business continuity, as IT failures can directly halt the company's operations and affect revenue and customer satisfaction.

Purpose of Our Disaster Recovery Plan

Our Disaster Recovery Plan (DRP) is designed to ensure that FalconPetro's IT-dependent operations can continue, even during major incidents. These incidents may include ransomware attacks, ERP or database failures, data center outages, or widespread POS system failures. The primary goal of the DRP is to maintain fuel supply, support financial operations, and allow fuel stations to continue serving customers while IT systems are being restored. The plan also protects the integrity, confidentiality, and availability of critical data.

Activation of the Disaster Recovery Plan

The DRP is activated whenever Tier-1 systems are affected, including:

- Karachi data center outage
- ERP system compromise or unavailability
- Supply chain and distribution database failure
- Widespread POS system failure across fuel stations

Once a critical incident is detected, its severity is assessed with the CISO and IT Operations teams. Upon confirmation, I formally authorize DRP activation and notify executive management, including the CEO and COO. This ensures that all teams are aware and ready to implement recovery procedures.

Recovery Strategy

Our recovery strategy focuses on minimizing operational disruption and restoring critical IT systems efficiently. The strategy includes the following steps:

1. **Shifting IT Operations:** Operations are moved from the primary data center in Karachi to the Disaster Recovery Site located in Lahore or cloud infrastructure. This site is

maintained in a ready state at all times.

2. **System Restoration from Backups:** Critical systems, including ERP, supply chain, and POS, are restored from verified backups. The integrity of backups is confirmed to ensure no corruption or malware is present.
3. **Reconnection of Fuel Stations and Depots:** Once systems are restored, fuel stations and depots are reconnected to ensure supply and transactions continue.
4. **Controlled Operations During Recovery:** Fuel supply continues in a limited and monitored capacity until full system restoration is complete.

This approach ensures that nationwide operations continue with minimal impact on customers and revenue.

Data Backup and Recovery Procedure

To prevent data loss, all critical systems are backed up daily and stored securely at the offsite DR site. During recovery:

- The latest clean backup is selected
- Backup integrity is verified to avoid corrupted or infected data
- ERP, supply chain, and POS data are restored and synchronized

This procedure protects financial and operational data, including pricing, inventory, and transaction records, maintaining business integrity.

Disaster Recovery Site (DR Site) Usage

Our DR site is constantly maintained and ready to handle emergencies. When activated:

- Servers and applications are powered and launched at the DR site
- Network routing and VPN access are updated to redirect operations
- Fuel stations and depots are connected to DR systems

By using the DR site, FalconPetro can continue nationwide operations even if the primary data center is completely offline.

System Testing Before Going Live

Recovered systems are never released immediately without proper testing. Before full-scale operations:

- ERP is tested for pricing, payroll, and procurement accuracy
- Supply chain systems are tested for tanker scheduling and distribution
- POS systems are tested with sample transactions to ensure correct payments and sales logging

Only after successful verification do we resume full operations, ensuring that restored systems are reliable and fully functional.

System Recovery Priority

We follow a priority-based recovery plan to restore critical systems first:

1. ERP System (SAP) – handles pricing, finance, and payroll
2. Supply Chain & Distribution Database – coordinates fuel delivery and tanker schedules
3. POS Systems & Payment Gateways – ensures customer transactions and revenue collection
4. Corporate Email & VPN – supports communication and coordination

This prioritization ensures that essential business operations are restored before secondary systems.

RTO (Recovery Time Objective) Plan

Definition: RTO is the maximum allowable time that a system can remain offline before it significantly impacts business operations.

FalconPetro RTO Plan:

1. **ERP System:** Must be restored within 4 hours to resume finance, payroll, and pricing operations.
2. **Supply Chain Database:** Must be restored within 2 hours to ensure fuel distribution schedules are maintained.
3. **POS & Payment Systems:** Must be restored within 1 hour to allow fuel sales and customer payments.
4. **Corporate Email & VPN:** Must be restored within 8 hours to maintain communication.

RTO Procedures:

- Immediately activate the DR site upon incident confirmation.
- Restore systems in priority order.
- Conduct rapid system testing to confirm operational readiness.
- Continuously monitor recovery progress to ensure each system meets its RTO.

Purpose: Ensures that downtime is minimized and critical business operations resume on time.

RPO (Recovery Point Objective) Plan

Definition: RPO is the maximum amount of data loss that is acceptable during a system failure or disaster.

1. **ERP System:** No more than 15 minutes of financial or operational data should be lost.
2. **Supply Chain Database:** Maximum 15 minutes of logistics and fuel schedule data may be lost.
3. **POS & Payment Systems:** Data must be near real-time to avoid losing customer transactions.
4. **Corporate Email & VPN:** Maximum 4 hours of email data may be lost.

RPO Procedures:

- Maintain frequent backups for all critical systems.
- Verify backup integrity before initiating restoration.
- Restore data from the latest clean backup to minimize losses.
- Ensure synchronization between DR site and primary site to prevent gaps.

Purpose: Ensures data loss is minimized and critical business information is preserved.

Fuel Station Operations During IT Outage

Even during a system outage, fuel stations continue operating in a limited capacity:

- POS offline mode is enabled for local transactions
- Cash-only payments are accepted
- Manual logs of sales and inventory are maintained
- Temporary fuel limits may be applied if required

This ensures that customers can still access fuel during emergencies.

Communication During Recovery

Effective communication is essential during recovery:

- I provide frequent updates to senior management on system restoration progress
- I coordinate with the COO to manage fuel station operations
- Security checks are performed with the CISO before systems go live
- IT teams receive clear instructions for recovery tasks

This ensures smooth, coordinated, and controlled recovery operations.

Returning to Normal Operations

Once the primary data center is restored:

- Data from the DR site is synchronized with the primary systems
- Systems are safely migrated back to Karachi
- DR site is returned to standby mode

This guarantees a seamless transition without data loss or operational interruption.

Post-Incident Review and Improvement

After the incident, a detailed review is conducted:

- Root cause analysis is performed to identify weaknesses
- Backup, monitoring, and recovery procedures are updated
- Lessons learned are incorporated into the DRP

This continuous improvement ensures that FalconPetro becomes more resilient and prepared for future incidents.

Lessons Learned

Importance of Risk Assessment

- Conducting a thorough risk analysis highlighted the vulnerabilities in network infrastructure, software, and employee practices.
- We learned that even minor oversights, such as weak passwords or outdated patches, can create significant security risks.

Role of Policies and Procedures

- Implementing structured policies (e.g., access control, data backup, incident response) is crucial for mitigating threats.
- Clear procedures ensure that employees know how to respond during security incidents, reducing downtime and potential losses.

Value of Network Segmentation

- Designing DMZs and internal networks taught us the benefits of isolating sensitive assets from public-facing systems.

- Segmentation reduces the attack surface and limits damage if one part of the network is compromised.

Need for Continuous Monitoring

- Security is not a one-time effort. Continuous monitoring, logging, and periodic audits are essential to detect and respond to threats promptly.

Team Collaboration and Practical Implementation

- Working together to design the IA plan reinforced real-world skills like network configuration, policy creation, and risk mitigation.
- Hands-on exercises helped connect theoretical concepts with practical security management.

8. Conclusion

The Information Assurance framework for FalconPetro Marketing Ltd. ensures the protection of critical IT, physical, and human assets, safeguarding the confidentiality, integrity, and availability of operations. Through risk assessment, robust policies, disaster recovery planning, and continuous monitoring, the company can prevent, respond to, and recover from cyber and operational incidents. This proactive approach enhances operational resilience, minimizes downtime, and maintains stakeholder confidence across nationwide fuel distribution and retail operations.